Cymbal Meet — Troubleshooting Guide: Device Performance

Diagnosing and Resolving Conference Room Device Issues

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1. Overview

Cymbal Meet conference room devices ("boxes") are purpose-built hardware units installed in meeting rooms. They provide high-definition video, spatial audio, and intelligent camera framing for in-room participants. When devices underperform, meeting quality degrades for all participants — both in-room and remote.

This guide covers the most common device performance issues, their root causes, diagnostic procedures, and resolution steps.

2. Performance Baselines

Understanding normal operating parameters is essential for identifying issues. Devices that consistently exceed these thresholds require investigation.

Normal Operating Parameters

Metric	Healthy Range	Warning Threshold	Critical Threshold
CPU Usage	20-45%	60-75%	>75%
Memory Usage	30-55%	65-80%	>80%
Network Latency	10-40ms	50-80ms	>80ms
Packet Loss	0.0-0.5%	0.5-2.0%	>2.0%
Video Quality Score	3.8-5.0	3.0-3.8	<3.0

Device Telemetry

Cymbal Meet devices report telemetry every 15 minutes during business hours (8 AM - 6 PM local time). Telemetry data includes CPU, memory, network, and video quality metrics. This data is available via the Cymbal Meet Admin Console and the BigQuery analytics pipeline.

3. Common Issues and Solutions

3.1 High Network Latency

Symptoms:

- Video freezing or stuttering
- Audio delay / echo
- Participants reporting "you're breaking up"

- Video quality score consistently below 3.5

Root Causes (in order of likelihood):

1. **Insufficient QoS (Quality of Service) Configuration**
    - The most common cause of latency issues, especially after office moves or network changes
    - Video traffic competes with bulk data transfers, cloud backups, and software updates
    - Impact: Latency spikes during business hours when network is congested
2. **Network Congestion**
    - Too many devices sharing insufficient bandwidth
    - Common when multiple conference rooms are in simultaneous use
3. **Routing Issues**
    - Traffic routing through unnecessary hops (VPN, proxy, or remote gateway)
    - Corporate proxy configurations that route video traffic through inspection appliances
4. **Physical Infrastructure**
    - Damaged or aging network cables
    - Switch port issues
    - WiFi interference (if device uses wireless backhaul)

Diagnostic Steps:

1. Check the Cymbal Meet Admin Console > Devices > Network Health for the affected room(s)
2. Compare latency patterns:
    - Consistent across all rooms → Network-wide issue (QoS, routing, ISP)
    - Isolated to specific rooms → Room-specific infrastructure issue
    - Time-of-day pattern → Congestion-related
3. Run a network diagnostic from the device: Settings > Network > Run Diagnostics
4. Check if a VPN or corporate proxy is in the traffic path: Settings > Network > Route Trace

Resolution Steps:

Cause	Action	Expected Impact
No QoS	Configure DSCP markings for Cymbal Meet traffic (EF for audio, AF41 for video). See Section 5 for detailed QoS configuration.	Latency drop of 40-70% within hours
Proxy routing	Exclude Cymbal Meet domains from proxy/VPN (*.cymbalmeet.com, media.cymbalmeet.com). Configure split tunneling.	Immediate latency improvement
Switch congestion	Upgrade to Gigabit switches, enable 802.1p priority queuing	Requires hardware change; 1-2 week timeline
Cable issues	Replace Cat5 with Cat6 cabling; test with cable certifier	Immediate after replacement

3.2 High Packet Loss

Symptoms:

- Audio cutting out (words/phrases missing)
- Video artifacts (blocky, pixelated video)
- Screen sharing showing stale content
- Video quality score drops below 3.0

Root Causes:

1. **Network Infrastructure Degradation**
    - Failing switches or routers dropping packets
    - Overloaded network buffers
    - Duplex mismatch on switch ports
2. **Bandwidth Saturation**
    - Total traffic exceeds available bandwidth
    - Common during peak hours or when large file transfers coincide with meetings
3. **Wireless Interference** (if using WiFi backhaul)
    - Competing WiFi networks on overlapping channels
    - Microwave ovens, Bluetooth devices, or other 2.4GHz interference
4. **ISP or WAN Issues**
    - Upstream packet loss beyond the organization's network boundary
    - Peering issues between ISP and Cymbal Meet's media servers

Diagnostic Steps:

1. Review packet loss trends in Admin Console > Devices > select device > Telemetry
2. If packet loss affects ALL devices across ALL rooms:
    - Issue is likely WAN/ISP-level or core network infrastructure
    - Contact ISP to check for upstream issues
    - Run traceroute from device to Cymbal Meet media servers
3. If packet loss affects devices in SPECIFIC rooms or buildings:
    - Issue is likely local infrastructure
    - Check switch port error counters
    - Verify cable integrity
4. If packet loss is intermittent and time-correlated:
    - Likely bandwidth saturation
    - Identify competing traffic using network monitoring tools

Resolution Steps:

1. **Immediate Mitigation:** Enable Forward Error Correction (FEC) on affected devices: Settings > Audio/Video > Enable FEC. This adds ~10% bandwidth overhead but recovers from 1-3% packet loss.
2. **Infrastructure Fix:** Address root cause per diagnostic findings (replace failing hardware, add bandwidth, fix QoS)
3. **Monitoring:** Set up alerts in Admin Console for packet loss exceeding 1% on any device

3.3 Low Video Quality Score

Symptoms:

- Blurry or low-resolution video
- Camera not switching to HD despite sufficient bandwidth
- Quality score consistently below 3.5 even with good network metrics

Root Causes:

1. **Bandwidth allocation below HD threshold**
    - Cymbal Meet requires 2.5 Mbps per device for 1080p video
    - If allocated bandwidth drops below 1.5 Mbps, quality is capped at 720p
    - Below 0.8 Mbps, quality drops to 480p
2. **CPU throttling**
    - Device CPU overheating, causing clock speed reduction
    - Running outdated firmware that has known performance regressions
3. **Camera hardware issues**
    - Lens obstruction or smudging
    - Camera sensor degradation (rare, but possible on units >3 years old)
4. **Lighting conditions**
    - Low ambient light forces camera to high ISO, introducing noise
    - Strong backlighting from windows causes exposure issues

Resolution Steps:

1. Verify bandwidth: Settings > Network > Bandwidth Test (expect >2.5 Mbps for HD)
2. Check firmware version: Settings > About > Firmware. Update if behind current release (see Section 4)
3. Check CPU thermals: Admin Console > Device > Hardware Health. If CPU temp >80C, check ventilation
4. Clean camera lens; verify room lighting meets minimum 300 lux at face level
5. If quality score remains low after above steps, contact Cymbal Meet Support for hardware diagnostic

3.4 High CPU / Memory Usage

Symptoms:

- Device UI is sluggish
- Slow to join meetings
- Features like intelligent framing or noise cancellation disabled automatically
- Device reboots unexpectedly

Root Causes:

1. **Outdated firmware** — older versions may have memory leaks or inefficient processing
2. **Too many background processes** — digital signage, room calendar display, and meeting room sensors running simultaneously
3. **Hardware aging** — devices older than 4 years may struggle with newer codec requirements

4. **Large meeting rooms** — devices in rooms with 20+ participants use more CPU for multi-stream processing

Resolution Steps:

1. Update firmware to the latest stable release (see Section 4)
2. Disable non-essential features: Settings > Features > disable Digital Signage, Room Sensors if not needed
3. Reboot the device (Settings > System > Restart) — resolves temporary memory leaks
4. If CPU consistently >70% after firmware update, consider hardware upgrade (contact your CSM for upgrade pricing)

4. Firmware Management

Current Firmware Versions

Device Model	Latest Stable	Latest Beta	End of Support
Cymbal Room 100	4.2.1	4.3.0-beta2	March 2027
Cymbal Room 200	4.2.1	4.3.0-beta2	June 2028
Cymbal Room 300	4.2.1	4.3.0-beta2	December 2029

Update Procedure

1. Schedule updates during non-business hours (updates take 10-15 minutes and reboot the device)
2. Admin Console > Devices > select devices > Schedule Firmware Update
3. Choose "Staged rollout" for organizations with 10+ devices — update 20% first, monitor for 48 hours, then continue
4. Verify post-update: check that video quality score and CPU/memory metrics are stable or improved

Known Firmware Issues

- **v4.1.x:** Memory leak in noise cancellation module. Devices reboot after 72 hours of continuous operation. Fixed in v4.2.0.
- **v4.0.x:** Packet loss measurement underreports by ~15%. Actual packet loss may be higher than telemetry shows. Fixed in v4.1.0.

5. Network Configuration — QoS Best Practices

DSCP Markings for Cymbal Meet Traffic

Traffic Type	DSCP Value	Per-Hop Behavior	Bandwidth Reservation
Audio (RTP)	46 (EF)	Expedited Forwarding	100 Kbps per device
Video (RTP)	34 (AF41)	Assured Forwarding	2.5 Mbps per device
Screen Share	26 (AF31)	Assured Forwarding	1.5 Mbps per device
Signaling (SIP/HTTPS)	24 (CS3)	Class Selector	50 Kbps per device

Minimum Bandwidth per Room

Room Capacity	Minimum Bandwidth	Recommended Bandwidth
1-6 people	5 Mbps	10 Mbps
7-15 people	10 Mbps	20 Mbps
16-30 people	15 Mbps	30 Mbps
30+ people	25 Mbps	50 Mbps

Firewall and Proxy Configuration

Required Allowlist:

- `*.cymbalmeet.com` — TCP 443 (signaling)
- `media.cymbalmeet.com` — UDP 10000-20000 (audio/video RTP)
- `telemetry.cymbalmeet.com` — TCP 443 (device telemetry)

Critical: Do NOT route media traffic (UDP) through a corporate proxy or SSL inspection appliance. This adds 30-80ms of latency and introduces packet loss. Use split tunneling or proxy bypass rules for Cymbal Meet media domains.

Network Health Check Checklist

- [ ] QoS DSCP markings configured on all switches between device and WAN uplink
- [ ] Firewall allows UDP 10000-20000 to media.cymbalmeet.com
- [ ] Proxy bypass configured for *.cymbalmeet.com
- [ ] Each meeting room has dedicated bandwidth allocation (not shared with general office WiFi)
- [ ] Switch ports for Cymbal Meet devices are on a dedicated VLAN with priority queuing
- [ ] Cable infrastructure is Cat6 or better
- [ ] WiFi access points (if used for backhaul) are on 5GHz with DFS channels

6. Hardware Replacement Criteria

A device should be replaced when:

1. **End of support date has passed** — no more firmware updates, security patches, or telemetry support
2. **CPU consistently >70%** after latest firmware update with minimal features enabled
3. **Video quality score consistently <3.0** despite good network conditions (verified by other devices on same network performing well)
4. **Device requires rebooting more than once per week** due to crashes or freezes
5. **Physical damage** — damaged camera, microphone array malfunction, speaker distortion

Contact your Customer Success Manager to discuss upgrade options and pricing. Cymbal Meet offers a trade-in program with 20% credit toward new devices when returning functional units.

7. Escalation Path

Level	Scope	Contact	SLA
Self-Service	Firmware updates, reboots, basic config	Admin Console	Immediate

L1 Support	Single-device issues, known issue workarounds	support@cymbalmeet.com	4 hours
L2 Support	Multi-device issues, network analysis, firmware bugs	Assigned CSM escalation	8 hours
L3 Engineering	Systemic issues, firmware defects, hardware recalls	CSM escalation to Engineering	24 hours

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Cymbal Meet Technical Support — For urgent device issues, contact support@cymbalmeet.com or call 1-800-CYMBAL-1 (24/7).